

Hypertension Guideline RAG

Evaluation Report: Retrieval & Generation Testing (200-Question Corpus)

AI Clinical Decision Support Hackathon

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Abstract

This document consolidates every retrieval, reranking, and generation experiment run for the hypertension-guideline RAG system on the 200-question benchmark set. Every table reports the metric values that were computed and stored on disk, and each table states the exact path of its output JSON file. A dedicated section explains the meaning and mathematical definition of every reported metric. The production configuration derived from these results is sparse BM25 on the deduplicated corpus ($k = 5$), with generation by Gemini Flash Lite at temperature 0.1.

1 Corpus and Test Sets

Corpus. Four guideline documents: WHO 2021 pharmacological treatment guideline, 2024 ESC guideline, USPSTF screening recommendation, and CDC Health E-Stats.

Test set (200 questions). All evaluation reported in this document uses the 200-question benchmark set (WHO 60 / ESC 90 / USPSTF 30 / CDC 20), stored as:

- `evaluation/test\_set\_200.json` — mapped to the original 813-chunk corpus.
- `evaluation/test\_set\_200\_dedup.json` — the same 200 questions remapped to canonical chunks of the deduplicated corpus (73 entries remapped).

Scoring. A retrieval is a hit if the expected `chunk_id` is returned *or* a returned chunk shares the expected `section_id`. Benchmark runner: `evaluation/benchmark\_200.py (-corpus original|dedup)`.

2 Evaluation Metrics and Their Meaning

This section defines every metric reported in the tables below.

2.1 Hit Rate @ k (**hit@ k**)

Meaning: the fraction of queries for which at least one *expected* chunk (or a chunk from the expected section) appears among the top- k retrieved chunks. It is a *recall*-type measure of how often the ground-truth evidence is captured.

$$\text{hit@}k = \frac{1}{N} \sum_{q=1}^N \mathbb{I}[\text{expected}(q) \in \text{top-}k(q)]$$

where N is the number of queries, $\mathbb{I}[\cdot]$ is the indicator function (1 if the condition holds, 0 otherwise), and $\text{top-}k(q)$ is the set of the k chunks returned for query q . Higher is better.

2.2 Precision @ k ($P@k$)

Meaning: the average fraction of the top- k retrieved chunks that are relevant (i.e. belong to the expected set). It measures how *noisy* the retrieved context is; high precision means the prompt context is tightly focused on the answer.

$$P@k = \frac{1}{N} \sum_{q=1}^N \frac{|\text{relevant}(q) \cap \text{top-}k(q)|}{k}$$

Higher is better, but precision and recall trade off against each other as k grows (including more chunks raises $\text{hit}@k$ but lowers $P@k$).

2.3 Mean Reciprocal Rank (MRR)

Meaning: the average of the reciprocal of the rank at which the first expected chunk appears. It rewards retrievers that place the correct evidence at the very top.

$$\text{MRR} = \frac{1}{N} \sum_{q=1}^N \frac{1}{\text{rank}(\text{first expected}(q))}$$

where $\text{rank}(\cdot)$ is 1 if the first expected chunk is the top result, 2 if it is second, and so on; if no expected chunk is retrieved, the term is 0. Range $[0, 1]$, higher is better.

2.4 Average Query Latency (ms)

Meaning: the mean wall-clock time taken by the retriever to answer a single query (including embedding API calls where applicable). Reported in milliseconds; lower is better. It captures the cost of the method (e.g. a cross-encoder reranker is far slower than plain BM25).

2.5 Correctness (0–5)

Meaning: an LLM judge (temperature 0.0) rates each generated answer against the expected answer on a 0–5 rubric, where 5 means the answer is fully correct and complete with respect to the guideline, and lower scores indicate partially correct, incomplete, or wrong answers. It measures *end-to-end answer quality* (retrieval + generation).

2.6 Faithfulness (0–5)

Meaning: an LLM judge rates whether every statement in the generated answer is *supported by the retrieved context*. 5 means the answer is fully grounded in the sources with no hallucination; lower scores indicate statements invented or contradicted by the context. It measures *grounding*, not factual completeness.

2.7 Citation Validity (0–1)

Meaning: the fraction of answers whose inline citations were judged valid against the retrieved sources by the LLM judge. 1.0 means every answer cited its sources correctly. Separately, a rule-based checker counts citations found / auto-valid / auto-invalid; discrepancies are mostly checker false-positives (child subsection cited vs parent section title retrieved).

3 200-Question Benchmark — Original Corpus (813 chunks)

All six retrieval modes on the full 200-question set: `evaluation/retrieval\_200\_results.json`.

Table 1: Retrieval benchmark, original corpus (813 chunks, $n = 200$).

Mode	hit@1	hit@3	hit@5	MRR	Latency (ms)
Sparse (BM25Okapi)	0.580	0.760	0.870	0.686	4.710
BM25 + CrossEncoder Rerank	0.575	0.770	0.880	0.688	1353.470
Dense (gemini-embedding-001)	0.345	0.560	0.650	0.456	31.570
Hybrid RRF ($c = 60$)	0.450	0.770	0.870	0.616	33.920
Hybrid Score ($a = 0.6$)	0.450	0.685	0.800	0.582	31.510
Hybrid RRF + CrossEncoder	0.525	0.755	0.890	0.658	1153.720

Source file: `evaluation/retrieval\_200\_results.json`

4 Corpus Deduplication (813 $\rightarrow$ 435 chunks)

Audit. Exact normalized-text grouping found 126 duplicate groups; 504 of 813 chunks involved; 378 redundant (46%). The pattern was the same text block assigned to many consecutive ESC sections (e.g. `sec_079–sec_087`).

Artifacts. Rebuilt corpus and remap tables (in `data/processed/`):

- `data/processed/validated\_chunks\_dedup.json` — 435 canonical chunks.
- `data/processed/embeddings\_dedup.json` — dense vectors for the dedup corpus.
- `data/processed/chunk\_remap\_dedup.json` — 378 duplicate $\rightarrow$ canonical mappings.

5 200-Question Benchmark — Deduplicated Corpus (435 chunks)

Same six modes on the dedup corpus: `evaluation/retrieval\_200\_dedup\_results.json`.

Table 2: Retrieval benchmark, deduplicated corpus (435 chunks, $n = 200$).

Mode	hit@1	hit@3	hit@5	MRR	Latency (ms)
Sparse (BM25Okapi)	0.755	0.890	0.945	0.826	2.000
BM25 + CrossEncoder Rerank	0.760	0.895	0.945	0.830	1027.970
Dense (gemini-embedding-001)	0.495	0.730	0.805	0.616	11.720
Hybrid RRF ($c = 60$)	0.650	0.890	0.955	0.772	14.140
Hybrid Score ($a = 0.6$)	0.630	0.870	0.955	0.753	13.230
Hybrid RRF + CrossEncoder	0.725	0.895	0.950	0.813	925.740

Source file: `evaluation/retrieval\_200\_dedup\_results.json`

Dedup impact (Δ vs original): BM25 hit@1 +17.5 pts, MRR +0.14; dense +15 pts; hybrid RRF +20 pts. Deduplication was the single biggest improvement.

6 Why Top- $k = 5$? (Retrieval Depth Justification)

The production retrieval depth ($k = 5$) is not the value suggested by a naive reading of the early pilot data, so this section documents the decision explicitly.

6.1 Initial pilot (10-question set) — ambiguous

A first tuning pass (`evaluation/tune\_k.py  $\rightarrow$  evaluation/top\_k\_tuning\_results.json`) tested $k \in \{1, 2, 3, 4, 5, 8, 10\}$ on a 10-question WHO pilot set using dense retrieval. There, hit rate saturated at 100% from $k = 2$, while precision fell as k grew, so the pilot alone would have suggested $k \in \{2, 3\}$

(its own weighted score tradeoff = $0.6 \cdot \text{hit} + 0.4 \cdot \text{precision}$ peaks at $k = 2$). This pilot is *not* the decision basis: only 10 questions, one document, dense-only, and trivially saturating recall.

6.2 200-question benchmark — decisive

At the 200-question scale, recall does **not** saturate at $k = 3$. The sparse BM25 numbers from Tables 1 and 2 give:

Table 3: BM25 hit@ k as a function of retrieval depth (200 questions).

k	Original (813 chunks)	Dedup (435 chunks)
1.000	0.580	0.755
3.000	0.760	0.890
5.000	0.870	0.945

Source file: `evaluation/retrieval\_200\_results.json`,
`evaluation/retrieval\_200\_dedup\_results.json`

Going from $k = 3$ to $k = 5$ recovers an additional **+11.0 pts** of recall on the original corpus and **+5.5 pts** on the production (dedup) corpus. That is, at $k = 3$ roughly 11 more of the 200 questions would arrive at the generator without their evidence chunk in context. The marginal cost is two extra chunks ($\approx 1,200$ tokens), a small price for a ≈ 5 – 11% recall gain; beyond $k = 5$ the precision@ k drops sharply ($62.5\% \rightarrow 54\% \rightarrow 35\%$ at $k = 4/5/8$) with almost no additional recall.

A k below 5 also risks missing multi-part recommendations (drug choice plus contraindications plus comorbidity guidance live in different chunks), while $k > 5$ dilutes the prompt context and increases cost. $k = 5$ is therefore the largest retrieval depth before marginal recall gain becomes negligible; on the dedup corpus $\approx 92\%$ of correct chunks are inside BM25 top-5. The residual misses (e.g. Q019, Q084, Q095) are not even in top-10, so no k recovers them — they are the queries for which the model correctly refuses rather than fabricates.

7 Medical Cross-Encoder (MedCPT) and Dense Variants

Clinical reranker/dense experiments on the dedup corpus (200 questions). Two files: `evaluation/medcpt\_results.` and `evaluation/medcpt\_stack\_results.json`.

Table 4: BM25 vs BM25 + MedCPT-Cross-Encoder (dedup corpus, $n = 200$).

Mode	hit@1	hit@3	hit@5	MRR	Latency (ms)
BM25 (baseline)	0.755	0.890	0.945	0.826	2.420
BM25 + MedCPT-CE	0.705	0.900	0.940	0.805	5624.810

Source file: `evaluation/medcpt\_results.json`

Table 5: MedCPT dense and hybrid stack (dedup corpus, $n = 200$).

Mode	hit@1	hit@3	hit@5	MRR	Latency (ms)
BM25 (baseline)	0.755	0.890	0.945	0.826	1.630
MedCPT dense (bi-encoder)	0.385	0.605	0.700	0.505	96.680
Hybrid RRF (BM25 + MedCPT dense)	0.610	0.830	0.905	0.727	106.950
Hybrid RRF + MedCPT-CE	0.700	0.905	0.955	0.804	5929.240

Source file: `evaluation/medcpt\_stack\_results.json`

Findings. Even the clinical cross-encoder (MedCPT, PubMedBERT-based) loses to plain BM25; root cause is 512-token truncation vs long guideline sections. `pritamdeka/S-PubMedBert-MS-MARCO` was

rejected as a reranker (bi-encoder, random classifier weights under `CrossEncoder`). `bge-reranker-base` is **incomplete** (evaluation timed out on CPU; weights downloaded and verified).

8 Answer Generation and Faithfulness (LLM)

Generation pipeline (`evaluation/eval\_generation.py`, resumable `generate/judge/report`) using BM25 top-5 on the dedup corpus as context, model `gemini-flash-lite-latest`, temperature 0.1, `max_output_tokens=1536`. Judge: temperature 0.0. All 200 records stored in: `evaluation/generation\_results.json`.

Table 6: End-to-end generation evaluation ($n = 200$).

Metric	Score
Correctness (0–5)	4.845
Faithfulness (0–5)	5.000
Citation validity (0–1, LLM judge)	0.990
Citations found	326.000
Citations auto-valid	279.000
Citations auto-invalid	47.000

Source file: `evaluation/generation\_results.json`

Table 7: Correctness by document (LLM judge).

Document	n	Correctness
WHO guideline	59.000	4.915
ESC guideline	94.000	4.798
USPSTF screening rec	29.000	5.000
CDC Health E-Stats	18.000	4.611

Source file: `evaluation/generation\_results.json`

Findings.

- Perfect grounding: zero hallucination across all 200 answers (faithfulness 5.0/5).
- All low-correctness cases are *retrieval* gaps (correct chunk absent from BM25 top-5); the model *refused cleanly* instead of inventing content.
- The 47 auto-checked “invalid” citations are mostly checker false-positives (child subsection cited vs parent section title retrieved; page offsets); the LLM judge scored 0.99.

9 Production Configuration

Final settings as deployed (`src/rag/rag\_pipeline.py`):

- Retriever: **sparse BM25** (`RETRIEVAL_MODE = "sparse"`).
- Corpus: `hypertension_guidelines_dedup` (435 chunks).
- `TOP_K = 5`; generation temperature 0.1, `max_output_tokens=1024`.
- Generation model: `gemini-flash-lite-latest`.

10 Key Conclusions

1. BM25 on the dedup corpus is the production retriever: 75.5% hit@1, MRR 0.826, ≈ 2 ms/query.

2. Deduplication (813→435) was the single biggest lever (+17.5 pts hit@1).
3. No dense, hybrid, or reranker variant beats plain BM25 on this corpus (2 embedding models, 3 cross-encoders, 6 fused/rerank combos tested).
4. End-to-end quality is strong: faithfulness 5.0/5, correctness 4.84/5; remaining errors trace to retrieval misses (≈ 11 queries), where the model correctly refuses.

11 Reproducibility

- All benchmark modes share one scorer in `evaluation/benchmark\_200.py` (`evaluate_mode`), configurable via `-corpus original|dedup`.
- Dense Gemini queries batch-embedded (100/request) to stay under the 100 req/min free-tier embedding quota.
- Dedup index rebuilt offline by reusing vectors from the original Qdrant collection (no re-embedding cost).
- Experiment log with per-query detail: `evaluation/EXPERIMENTS.md`.